

HEALTHCARE DATA CLIENT

END-TO-END Project Report

Healthcare Data Quality Automation and Exception Reporting

Project Summary	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Data Scale	Approximately 250,000 records per batch
Observed Exception Rate	2%
Review Turnaround	Reduced from 6 hours to 2.5 hours per file
Tools	Python, SQL, Excel, Power Query and Power BI

Report basis: This report consolidates confirmed engagement facts and clearly identified portfolio reconstruction elements. It does not introduce confidential healthcare data, unsupported clinical rules or fabricated historical results.

1. Executive Summary

The Healthcare Data Client engagement focused on improving the review of high-volume healthcare data through automated data-quality validation and exception reporting. The solution processed approximately 250,000 records per batch across healthcare data domains including claims, patients, appointments, providers, billing, transactions, clinical records and insurance data.

Automated checks covered duplicates, missing values, invalid dates, incorrect formats, invalid identifiers, out-of-range values, inconsistent codes and cross-field mismatches. Records failing one or more applicable rules were flagged, separated for review and assigned reason information. The engagement surfaced a 2% record-level exception rate and reduced review turnaround from approximately 6 hours to 2.5 hours per file.

Headline impact: 3.5 hours less review turnaround per file, equivalent to an approximately 58.3% reduction derived from the confirmed before- and-after values.

2. Business Problem

Challenge	Business Need
High record volume	Apply consistent validation across approximately 250,000 records per batch.
Multiple data-quality conditions	Detect different types of record-level validation failures systematically.
Exception identification	Separate records requiring attention from the processed population.
Review context	Provide reason information so analysts can understand why a record was flagged.
Turnaround	Reduce the time required to review each file.
Reporting visibility	Provide Power BI visibility into governed data-quality and exception results.

3. Project Objectives

Objective	Outcome / Delivery
Automate data-quality checks	Repeatable validation across the confirmed rule categories.
Improve exception visibility	Record-level exception flagging and separate review output.
Support targeted review	Reason codes or equivalent failure explanations attached to exceptions.
Maintain analytical integrity	Separate record-level Exception Records from rule-level Validation Failures.
Provide reporting visibility	Power BI reporting based on governed validation output.
Reduce review turnaround	Confirmed reduction from 6 hours to 2.5 hours per file.

4. Scope & Data Landscape

Area	Confirmed Scope
Data domains	Claims, patients, appointments, providers, billing, transactions, clinical records and insurance data
Sources	Excel, CSV and SQL/database
Tools	Python, SQL, Excel, Power Query and Power BI
Primary stakeholder	Healthcare Analyst
Processing scale	Approximately 250,000 records per batch
Public confidentiality	No PHI, patient identifiers, provider identifiers or confidential client values included

5. Requirements & Business Rules

Requirements centered on high-volume ingestion, repeatable validation, exception classification, reason-coded outputs, traceability and Power BI reporting. The central record-level rule was that a record failing one or more applicable validation checks is treated as an exception.

Governed Rule	Definition
Exception Record	Distinct processed record failing one or more applicable validation rules.
Exception Rate	Exception Records / Records Processed x 100.
Exception Output	Separate file, table or report containing records requiring review.
Reason Traceability	Failed records carry a reason code or equivalent failure explanation.
Multiple Failures	One record may have multiple rule-level failures but remains one record-level exception.

Observed rate: The 2% Exception Rate is a confirmed analytical result. It is not presented as a target, tolerance or SLA.

6. Data Preparation & Validation

Stage	Approach
Source intake	Establish the governed batch population and source context.
Profiling	Assess structure, apparent types, missing values and formatting conditions.
Preparation	Standardize only where required for reliable validation while preserving traceability.
Rule execution	Apply documented validation logic to applicable fields.
Exception classification	Flag failed records and capture reason information.
Reconciliation	Reconcile processed records, exception records and rule-level failures.
Reporting preparation	Prepare governed outputs for Power BI.

7. Data Quality Rule Coverage

Rule Family	Quality Dimension	Purpose
Duplicate Detection	Uniqueness	Identify records meeting the documented duplicate definition.
Missing-Value Check	Completeness	Identify applicable null, blank or missing values.
Date Validation	Validity	Identify dates failing the documented date rule.
Format Validation	Conformance	Identify values not matching the documented expected format.
Identifier Validation	Validity	Identify IDs failing the documented identifier rule.
Range Validation	Validity	Identify values outside documented bounds.
Code Consistency	Consistency	Identify values inconsistent with documented codes or mappings.
Cross-Field Validation	Integrity	Identify violations of documented relationships between fields.

8. Automated Validation Workflow

Step	Process
1	Ingest
2	Validate
3	Apply Data Quality Rules
4	Flag Exceptions
5	Assign Reason Codes

6	Generate Exception Output / Report
7	Review
Historical boundary: The high-level workflow is confirmed. Exact orchestration, scheduling, infrastructure and original scripts are not retained.	

9. Exception Analysis

Analysis Area	Governed Treatment
Record-level exception	Count each affected processed record once.
Rule-level failure	Allow multiple failures for the same record.
Reason analysis	Use governed reason code or validation-rule detail where available.
Source traceability	Maintain linkage to the affected source record.
Domain analysis	Use healthcare-domain classification only where explicitly supported by source data.
Historical distribution	Do not invent per-rule, per-domain or reason-code frequencies.

10. Power BI Solution

Power BI provided reporting visibility for the validation and exception process. Because the exact historical page names and layouts are not retained, the portfolio solution organizes reporting into three evidence-safe views.

Portfolio View	Purpose
Data Quality Executive Overview	Summarize processed volume, Exception Rate, validation scope and review-turnaround context.
Exception Analysis	Support investigation by governed reason or validation rule and controlled record-level review.
Validation Performance	Explain validation coverage, record-versus-failure grain and reconciliation controls.

11. Data Model & KPI Logic

Model Area	Control
Processed-record grain	Supports Records Processed, Exception Records and record-level Exception Rate.
Validation-failure grain	Supports rule-level failure and reason analysis.
Relationship control	Rule-level rows must not multiply record-level KPIs.
Exception Rate	Distinct Exception Records / Records Processed x 100.
Review Time Reduction	6.0 hours - 2.5 hours = 3.5 hours per file.
Turnaround Reduction %	3.5 / 6.0 x 100 = approximately 58.3%.

12. Testing & Quality Assurance

Test Area	Control
Input reconciliation	Processed population reconciles to the governed batch.
Rule testing	Each implemented rule produces expected pass/fail behaviour.
Exception handling	Failed records are flagged and reason-coded.
Multi-failure handling	One record with multiple failures remains one Exception Record.
KPI reconciliation	Exception Rate uses governed record-level populations.
Power BI validation	Dashboard measures reconcile to validation outputs.
Batch-scale reference	Workflow supports the confirmed approximate 250,000-record batch scale.

Testing boundary: Exact historical test logs, defect counts, formal UAT approvals and production performance SLAs are not retained.

13. Findings

Finding	Evidence-Safe Interpretation
Approximately 250,000 records per batch	The engagement operated at substantial healthcare data volume.
2% observed Exception Rate	A distinct subset of processed records required review under the implemented validation logic.
Reason-coded exception output	Analysts could understand why flagged records required attention.
6 hours to 2.5 hours per file	Automated validation and targeted exception review materially reduced review turnaround.
Approximately 58.3% reduction	Derived turnaround reduction only, not a productivity, cost, clinical or compliance metric.

14. Recommendations

Recommendation	Rationale	Status
Maintain a governed validation rule catalogue	Keeps rule logic, parameters and reason classifications testable and repeatable.	Portfolio recommendation
Separate record and failure grains	Prevents double-counting in Exception Rate while preserving detailed failure analysis.	Portfolio recommendation
Reconcile every processed batch	Protects KPI and reporting integrity.	Portfolio recommendation
Retain source and batch traceability	Supports controlled investigation of exceptions.	Portfolio recommendation
Regression-test rule changes	Reduces risk when formats, ranges, codes or relationships change.	Portfolio recommendation
Keep Power BI aligned with validation outputs	Prevents reporting logic from diverging from automation logic.	Portfolio recommendation

15. Business Impact

Measure	Result
Processing scale	Approximately 250,000 records per batch
Observed Exception Rate	2%
Baseline review turnaround	6 hours per file
Post-solution review turnaround	2.5 hours per file
Time reduction	3.5 hours per file
Derived turnaround reduction	Approximately 58.3%

16. Role Contribution

As Data Analyst, the engagement combined healthcare data review, validation-rule implementation, exception analysis, automation, reporting and quality assurance. Requirements, business rules, testing and documentation were also part of the project scope. Tools used across the engagement included Python, SQL, Excel, Power Query and Power BI.

17. Evidence & Confidentiality Boundaries

Supported	Not Presented as Historical Fact
Confirmed data domains, sources, tools and stakeholder	Exact source schema, table names and column names
Eight confirmed validation categories	Unretained healthcare-specific thresholds or clinical code logic
Approximately 250,000 records per batch	Exact per-rule or per-domain exception counts
2% observed Exception Rate	2% as an SLA or acceptable-error threshold
6 hours to 2.5 hours per file	Detailed composition of remaining review time
Exception flagging and reason codes	Formal UAT sign-off, defect counts or production architecture

18. Project Conclusion

Conclusion: The project demonstrates a complete data-quality lifecycle: high-volume healthcare data intake, governed validation, traceable exception handling, targeted review, Power BI reporting and measurable turnaround improvement. Its strongest defensible outcome is the reduction of review turnaround from 6 hours to 2.5 hours per file while surfacing a 2% record-level exception rate across approximately 250,000 records per batch.